

English Indices of Multiple Deprivation Analysis

Group Investigation

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Author's Note

This investigation was completed as part of a collaborative MSc Data Science group project. The sections presented in this portfolio represent my own contributions to the investigation (Questions 1, 2 and 5). Questions 3 and 4 were completed by other members of the project and are included to preserve the context of the overall investigation.

1. Data Preparation

1.1 Lookup table

To support the geographic analysis, a lookup table was constructed to connect local authority districts and counties with their corresponding English regions. Two Office for National Statistics lookup datasets were imported and joined using the common local authority district identifier, LAD24CD. The required county and regional identifiers and names were then retained to create a concise reference table for subsequent data integration.

The following R code was used to construct the geographic lookup table:

```
library(dplyr)

cty =
read_csv('Local_Authority_District_to_County_(December_2024)_Lookup_in_EN.csv')

rgn =
read_csv('Local_Authority_District_to_Region_(December_2024)_Lookup_in_EN.csv')

cty
rgn

lookup_table = cty %>%
  left_join(rgn, by = 'LAD24CD') %>%
  select(LAD24CD, CTY24CD, CTY24NM, RGN24CD, RGN24NM)

lookup_table
```

The resulting lookup table provides a structured mapping between local authority districts, counties and regions. This enabled missing regional information within the IMD dataset to be recovered using the available geographic identifiers.

1.2 Augment the IMD 2025 dataset

The original IMD dataset contained missing regional values for a number of observations. To improve the completeness of the dataset, the geographic lookup table was joined to the IMD data using the county identifier. The `coalesce()` function was then applied to retain existing regional values where available and replace missing values with the corresponding regional name from the lookup table.

The following R code was used to augment the IMD dataset and resolve missing regional information:

```
imd_group = read_csv("imd2025_group.csv")
imd_group
aug_imd = imd_group %>%
  left_join(lookup_table, by = c("LAD24CD" = "CTY24CD")) %>%
  mutate(Region = coalesce(Region, RGN24NM)) %>%
  distinct()

aug_imd
write_csv(aug_imd, "Augmented_DataSet.csv")
```

Following the join, validation checks were performed to inspect the resulting variables and confirm whether any regional values remained missing.

```
sum(is.na(aug_imd$RGN24NM))
names(imd_group)
names(lookup_table)
```

These checks were used to confirm that the lookup variables had been incorporated correctly and to identify any records that might require further investigation. Removing duplicate rows with `distinct()` also ensured that the join did not unintentionally produce repeated observations.

1.3 Summary table

Following augmentation, the dataset was grouped by region to examine the geographic distribution of local authority districts. A summary table was produced showing the number of districts represented within each English region.

The following R code was used to calculate the number of districts within each region:

```
region_summary = aug_imd %>%
  group_by(Region) %>%
  summarise(n_districts = n()) %>%
  arrange(Region)
region_summary
```

The resulting regional summary is presented below.

```
# A tibble: 9 × 2
  Region                n_districts
  <chr>                 <int>
1 East Midlands         35
2 East of England       45
3 London                33
4 North East            12
5 North West            35
6 South East            64
7 South West            27
8 West Midlands         30
9 Yorkshire and The Humber 15
```

The summary confirms that the number of districts varies considerably between regions. The South East contains the largest number of districts, with 64 observations, while the North East contains the fewest, with 12. This imbalance is important when interpreting regional comparisons because regions with more districts contribute a larger number of observations to the analysis. Consequently, differences in sample size should be considered when comparing distributions and identifying wider geographic patterns.

To investigate relationships between the seven IMD deprivation domains, a scatter matrix was produced using the GGally package. The scatter matrix provides pairwise visualisations of each variable alongside their corresponding correlation coefficients, enabling strong positive and negative relationships to be identified prior to more advanced statistical analysis.

The following R code was used to generate a scatter matrix for the seven IMD domains.

```
library(GGally)

domains = new_table %>%
  select(Income, Employment, Education, Health, Crime, Barriers,
  Living)

ggpairs(domains)
```

The resulting scatter matrix is presented in Figure 1.

Figure 1

Figure 1. Scatter matrix illustrating pairwise relationships between the seven IMD deprivation domains. The plots were used to identify potential correlations and underlying relationships prior to further statistical analysis.

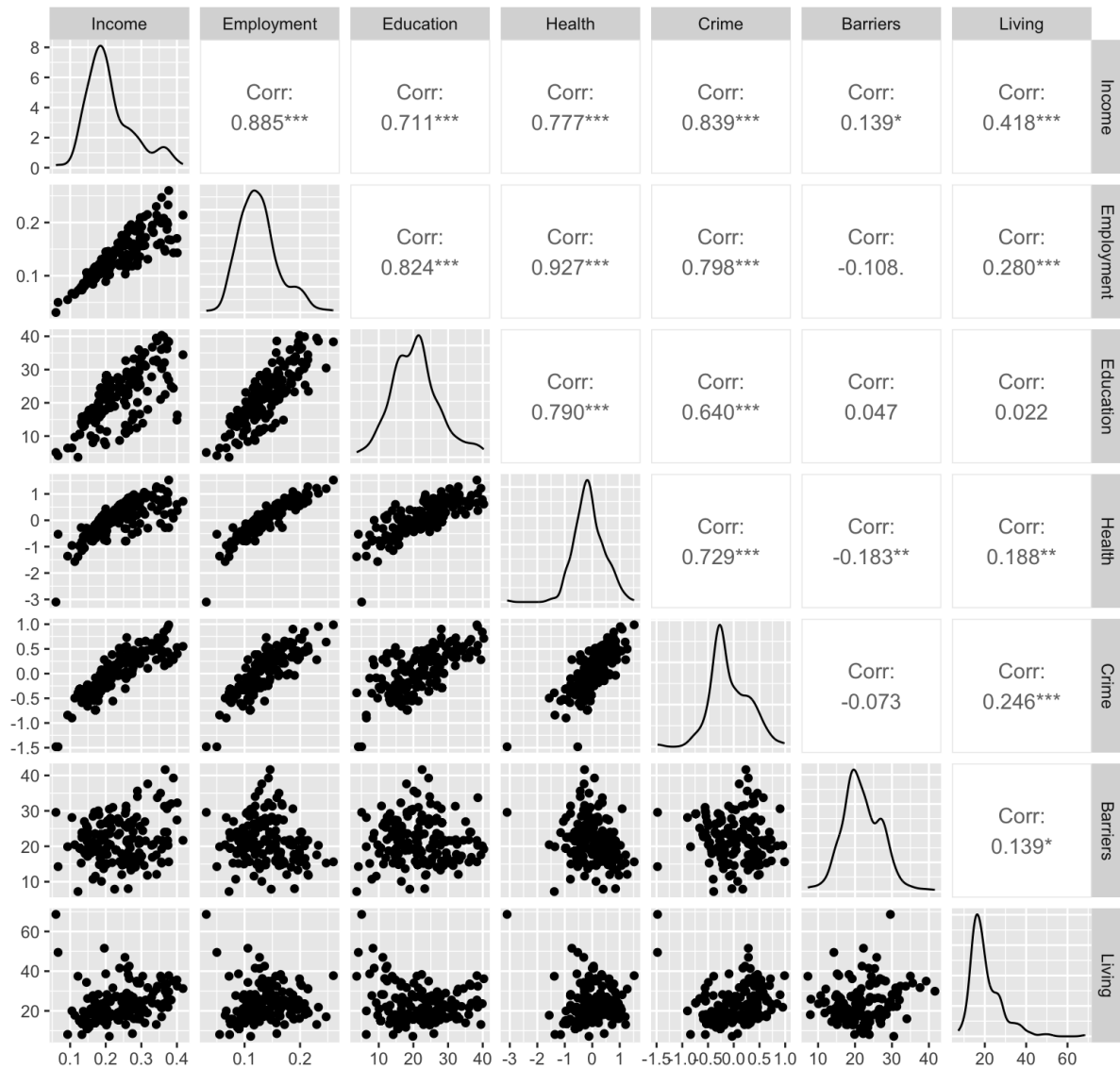


Figure 1 demonstrates several strong positive correlations between the deprivation domains, indicating that many aspects of deprivation tend to occur simultaneously rather than in isolation. In particular, Employment, Income, Education and Health exhibit especially strong relationships, suggesting that districts experiencing deprivation in one of these domains are likely to experience deprivation in the others.

The strongest observed relationship occurs between Employment and Health (correlation coefficient = 0.927), indicating that areas experiencing higher employment deprivation also tend to experience poorer health outcomes. This relationship is plausible because

unemployment is often associated with reduced income, poorer living conditions and decreased access to resources that support good physical and mental health.

Similarly, the strong positive relationship between Employment and Income reflects the close connection between labour market participation and financial wellbeing. Districts with higher employment deprivation are more likely to experience income deprivation because employment is a primary source of household income. Likewise, strong correlations involving Education suggest that limited educational opportunities may reduce future employment prospects, contributing indirectly to income deprivation.

The positive relationships between Crime, Income and Employment also indicate that socioeconomic deprivation is associated with increased crime deprivation. Although correlation does not imply causation, these findings suggest that areas experiencing financial hardship may also face broader social challenges that contribute to higher crime levels. Similarly, the relationship between Crime and Health may reflect the wider impact of deprived neighbourhoods on both physical and mental wellbeing.

Overall, the scatter matrix demonstrates that the IMD domains are highly interconnected, supporting the use of multivariate statistical techniques such as Principal Component Analysis (PCA) and hierarchical clustering in the subsequent stages of the investigation.

(this question was completed by Mia Emanuele and Contributor 2)

2. Exploratory Data Analysis

2.1 North and South dataset

To investigate potential regional differences in deprivation, a subset of the augmented IMD dataset was created containing only districts located in the North and South of England. Districts belonging to the Midlands were excluded to provide a clearer comparison between the two geographical regions. This simplified dataset enabled differences across the seven deprivation domains to be explored using comparative boxplots.

2.2 Quantitative variables to predict North vs South

The following R code was used to generate boxplots comparing the distribution of each IMD deprivation domain between the North and South of England.

```
north_south = read_csv("Augmented_DataSet_north_south.csv")

ggplot(north_south, aes(x = Region, y = Education, fill = Region)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Education Deprivation by Region", y = "Education
Score", x = "") +
  theme_minimal() +
  theme(legend.position = "none")

ggplot(north_south, aes(x = Region, y = Income, fill = Region)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Income Deprivation by Region", y = "Income Score", x =
"") +
  theme_minimal() +
  theme(legend.position = "none")

ggplot(north_south, aes(x = Region, y = Employment, fill = Region)) +
  geom_boxplot(alpha = 0.7) +
```

```

    labs(title = "Employment Deprivation by Region", y = "Employment
Score", x = "") +
    theme_minimal() +
    theme(legend.position = "none")

ggplot(north_south, aes(x = Region, y = Health, fill = Region)) +
    geom_boxplot(alpha = 0.7) +
    labs(title = "Health Deprivation by Region", y = "Health Score", x =
"") +
    theme_minimal() +
    theme(legend.position = "none")

ggplot(north_south, aes(x = Region, y = Crime, fill = Region)) +
    geom_boxplot(alpha = 0.7) +
    labs(title = "Crime Deprivation by Region", y = "Crime Score", x =
"") +
    theme_minimal() +
    theme(legend.position = "none")

ggplot(north_south, aes(x = Region, y = Barriers, fill = Region)) +
    geom_boxplot(alpha = 0.7) +
    labs(title = "Barriers Deprivation by Region", y = "Barriers Score",
x = "") +
    theme_minimal() +
    theme(legend.position = "none")

ggplot(north_south, aes(x = Region, y =Living, fill = Region)) +
    geom_boxplot(alpha = 0.7) +
    labs(title = "Living Deprivation by Region", y = "Living Score", x =
"") +
    theme_minimal() +
    theme(legend.position = "none")

```

Seven boxplots were initially produced to compare the distribution of each IMD deprivation domain between the North and South of England. Following visual inspection, the Health and Education domains were selected for further analysis because they exhibited the clearest regional separation, with minimal overlap between the northern and southern distributions. Consequently, these two variables were considered the most informative for distinguishing between the two geographical regions.

Figure 2

Distribution of Health deprivation scores in the North and South of England.

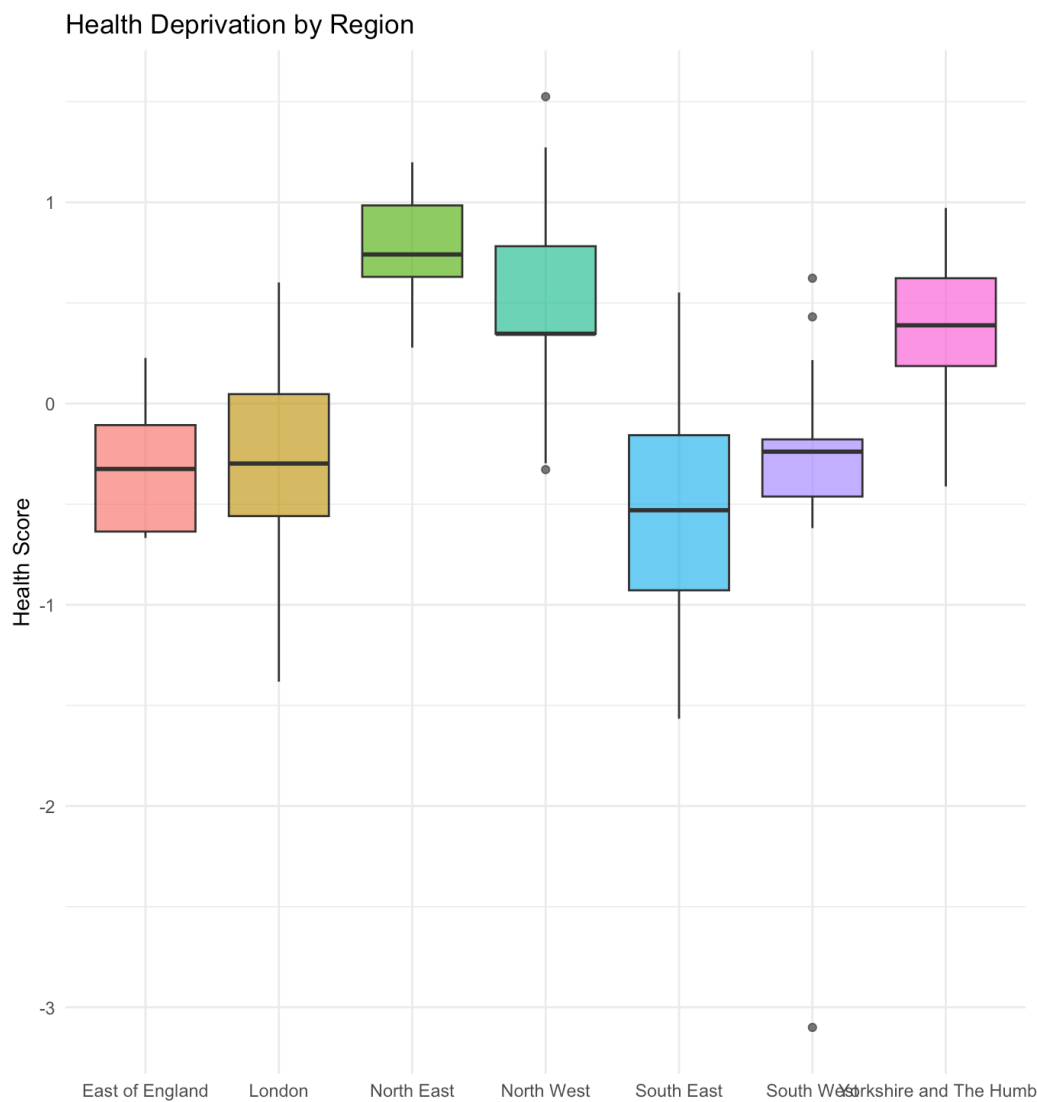
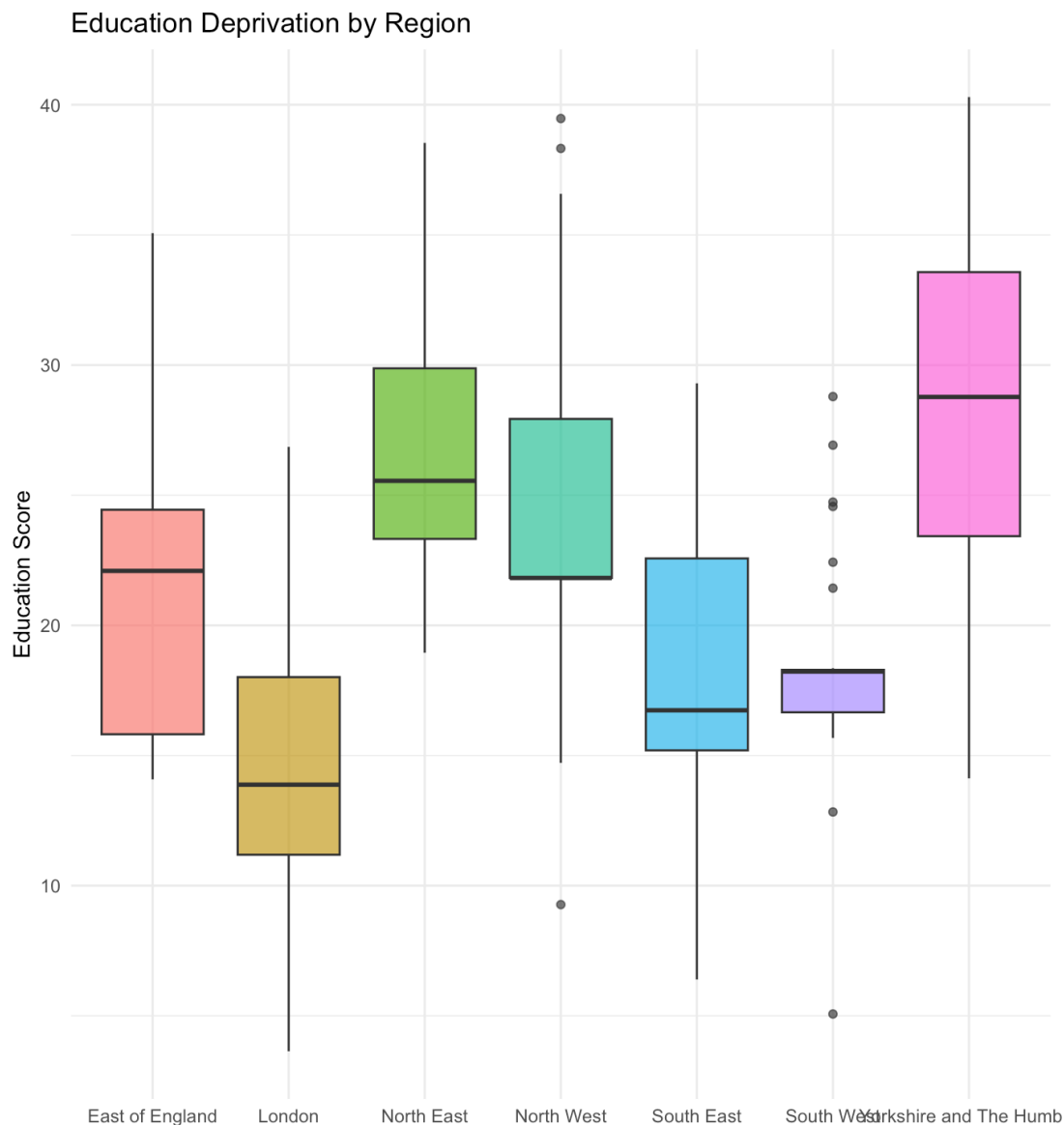


Figure 3

Distribution of Education deprivation scores in the North and South of England.



Figures 2 and 3 demonstrate noticeable regional differences in both Health and Education deprivation. In each case, districts in the North of England generally exhibit higher deprivation scores than those in the South, suggesting that these forms of deprivation are more prevalent in northern districts. Although the distributions overlap, the higher median values observed for the North indicate a persistent regional disparity.

The boxplots also identify several districts that deviate substantially from the overall pattern. In the Health boxplot, districts such as Cheshire East and the Isles of Scilly display comparatively low health deprivation despite being located in regions where different

outcomes might be expected. Conversely, Torbay exhibits relatively high health deprivation despite being situated in the South of England. These observations suggest that regional trends do not fully explain variation between individual districts and that local socioeconomic factors may also influence deprivation.

A notable finding is that Blackpool, Plymouth, Torbay and the Isles of Scilly appear as outliers in both the Health and Education boxplots. The presence of the same districts as outliers across multiple deprivation domains suggests that these observations are less likely to be the result of data entry errors and more likely to represent genuine variation within the dataset. Consequently, these districts warrant further investigation to better understand the factors contributing to their distinctive deprivation profiles.

Overall, the boxplots demonstrate that both Health and Education deprivation vary geographically across England while also highlighting districts whose deprivation patterns differ markedly from the broader regional trend. These findings provide a useful foundation for the multivariate analyses presented later in the investigation.

2.3 Districts difficult to predict

Following the identification of statistical outliers in the Health and Education deprivation domains, the following R code was used to detect observations lying beyond 1.5 times the interquartile range (IQR) within each region. These observations were subsequently matched to their corresponding districts for further investigation.

```
health_stats = north_south %>%
  group_by(Region) %>%
  summarise(
    Q1 = quantile(Health, 0.25),
    Q3 = quantile(Health, 0.75),
    IQR = Q3 - Q1)
health_outliers = north_south %>%
  left_join(health_stats, by = "Region") %>%
  filter(Health < (Q1 - 1.5*IQR) | Health > (Q3 + 1.5*IQR)) %>%
  select(Region, Health)
health_outliers
```

```

education_stats = north_south %>%
  group_by(Region) %>%
  summarise(
    Q1 = quantile(Education, 0.25),
    Q3 = quantile(Education, 0.75),
    IQR = Q3 - Q1)
education_outliers = north_south %>%
  left_join(education_stats, by = "Region") %>%
  filter(Education < (Q1 - 1.5*IQR) | Education > (Q3 + 1.5*IQR)) %>%
  select(Region, Education)
education_outliers

```

The identified outliers for the Health and Education deprivation domains are presented below.

```

# A tibble: 5 × 2
  Region      Health
  <chr>      <dbl>
1 North West  1.52
2 South West  0.431
3 South West  0.623
4 North West -0.328
5 South West -3.1

```

```

# A tibble: 11 × 2
  Region      Education
  <chr>      <dbl>
1 North West  38.3
2 South West  12.8
3 South West  24.6
4 South West  26.9

```

5	South West	28.8
6	South West	24.7
7	South West	21.4
8	South West	5.07
9	South West	22.4
10	North West	9.27
11	North West	39.5

The Health deprivation analysis identified Blackpool, Plymouth, Torbay, Cheshire East and the Isles of Scilly as statistical outliers. Similarly, the Education deprivation analysis identified Blackpool, Bath and North East Somerset, Bristol, Plymouth, Torbay, Swindon, Cornwall, the Isles of Scilly, Somerset, Trafford and Knowsley as outlying districts. These districts differ substantially from the typical deprivation patterns observed within their respective regions and therefore warrant further investigation.

Several of these districts appear difficult to classify because their deprivation profiles do not follow the general regional trend. For example, Trafford, a district located in the North of England, exhibits a comparatively low Education deprivation score despite northern districts generally displaying higher levels of educational deprivation. Conversely, Torbay, located in the South of England, demonstrates a relatively high Education deprivation score, making its deprivation profile more characteristic of northern districts. These observations suggest that regional location alone does not fully explain deprivation patterns across England.

A similar pattern is observed within the Health deprivation domain. Cheshire East and the Isles of Scilly exhibit comparatively low Health deprivation despite belonging to regions where higher deprivation might be expected, whereas Torbay displays relatively high Health deprivation for a southern district. These contrasting observations highlight the complexity of socioeconomic deprivation and suggest that local characteristics may influence deprivation independently of broader regional trends.

Particularly noteworthy is the fact that Blackpool, Plymouth, Torbay and the Isles of Scilly appear as outliers in both the Health and Education analyses. While isolated outliers can sometimes arise from data entry or measurement errors, observations that are consistently identified as outliers across multiple variables are more likely to represent genuine underlying variation within the dataset. Consequently, these districts should be considered priority candidates for further investigation to better understand the socioeconomic factors contributing to their distinctive deprivation profiles.

(this question was completed by Mia Emanuele and Contributor 1)

3. Principal Component Analysis (*team contribution*)

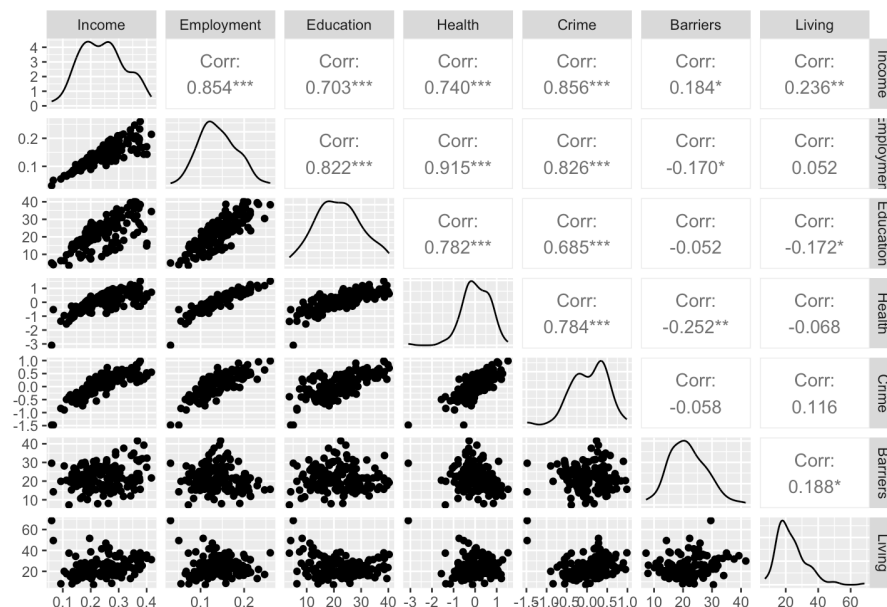
3a) Use R to carry out Principal Component Analysis (PCA) on the dataset. You may find the R package "factoextra" useful.

Principal component Analysis employed to explore the underlying structure of deprivation across districts using the seven IMD domains. PCA is a widely used multivariate technique that reduces dimensionality by transforming correlated variables into a smaller set of uncorrelated principal components, while retaining the majority of the original variance. This approach facilitates interpretation by summarising complex, multi-domain deprivation patterns into a limited number of meaningful components.

Prior to PCA, all IMD domains were standardised to ensure equal contribution to the analysis and to prevent domains with larger variances from disproportionately influencing the results. The resulting principal components are ordered according to the amount of variance they explain, with the first component capturing the largest share of overall variation and subsequent components accounting for progressively smaller proportions.

Figure 4

Scatter matrix



Code:

```
imd_scaled <- scale(imd_domains)
```



```

pca_full <- prcomp(imd_scaled, center = TRUE, scale. = TRUE)

cat("\nPCA scree plot...\n")
print(fviz_eig(pca_full, addlabels = TRUE))

cat("\nPCA variable loadings plot...\n")
print(fviz_pca_var(pca_full, repel = TRUE))

cat("\nPCA biplot PC1 vs PC2 coloured by Region...\n")
print(fviz_pca_biplot(pca_full, geom.ind = "point", habillage =
imd_aug$Region, repel = TRUE))

cat("\nPCA biplot PC2 vs PC3 coloured by Region...\n")
print(fviz_pca_biplot(pca_full, axes = c(2,3), geom.ind = "point",
habillage = imd_aug$Region, repel = TRUE))

```

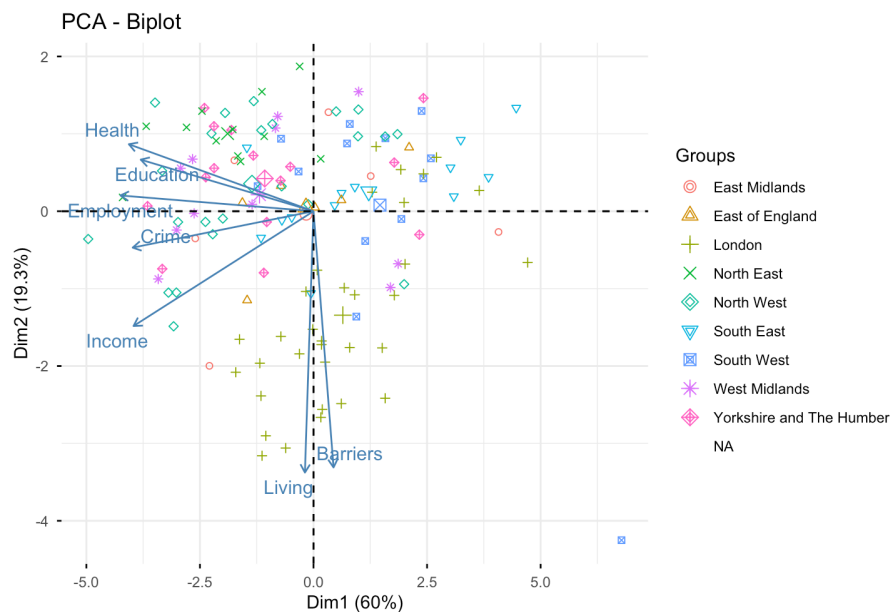
Interpretation of PC1:

The first principal component (PC1) accounts for the largest variance in the IMD dataset. PC1 is considered a common dimension of deprivation. The loadings plot shows that Income, Employment, Education, and Health have a strong and positive impact on PC1. Therefore, if a locality has a high score in PC1, it will have a higher level of deprivation in all four dimensions. As these dimensions have a high degree of correlation among them, PC1 is considered a good factor for capturing deprivation intensity. Additionally, it can be observed from the PCA biplot that a majority of localities are located along this dimension.

Interpretation of PC2:

The second principal component, PC2, reflects the variation not explained by Overall Deprivation. PC2 is driven mainly by Crime and Barriers to Housing and Services, with weaker or opposing associations with Income and Employment. PC2 differentiates neighborhoods in which Overall Deprivation is more a function of service access and crime factors rather than purely economic circumstances. In explaining PC1/PC2 biplot, neighborhoods are differentiated in terms of PC2, emphasizing their divergent structures in terms of deprivation despite similarities in Overall Deprivation.

Figure 5
PCA Biplot

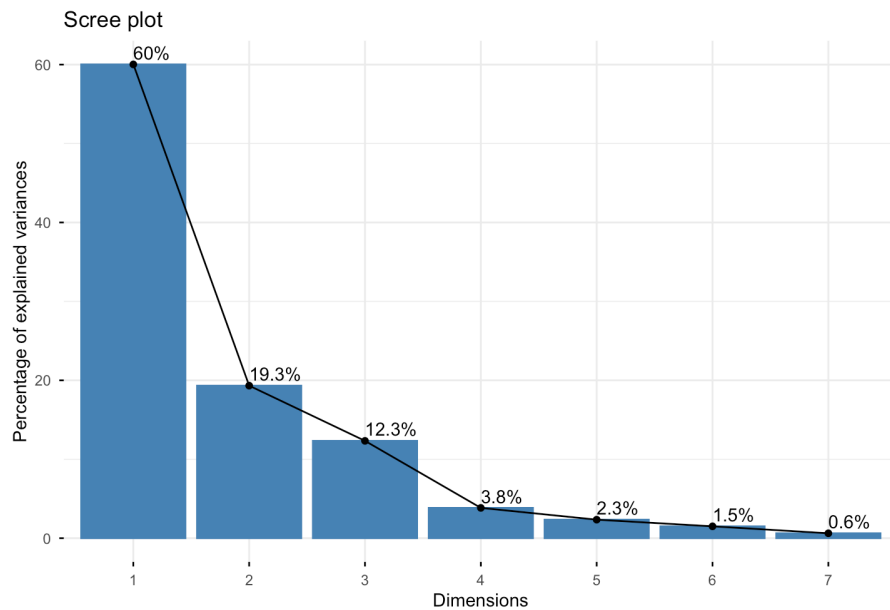


Intepretation of PC3:

The third principal component, PC3, explains a smaller but still meaningful proportion of variance. PC3 is associated mainly with the Living Environment domain, which suggests that environmental deprivation varies independently from socio-economic deprivation. The PC2–PC3 biplot presents further separation between districts and shows that the environmental conditions provide further differentiation beyond income, employment, and health-related facts.

Figure 6

Scree plot



Output:

Console PCA scree plot...

PCA variable loadings plot...PCA biplot PC1 vs PC2 coloured by Region...Graphical outputs (Plots tab)

- Scree Plot
 - Bar chart showing eigenvalues
 - X-axis: PC1, PC2, PC3, ...
 - Y-axis: Percentage of variance explained
 - Cumulative variance line shown
- PCA Variable Loadings Plot
 - Arrows representing IMD domains
 - Direction = correlation with PCs
 - Length = strength of contribution
- PCA Biplot (PC1 vs PC2)
 - Points = observations
 - Arrows = IMD variables
 - Colours = Regions
 - Ellipses around regions

PC1	PC2	PC3	Income	0.42	-0.18	0.07	Employment
0.40	-0.22	0.05	Education	0.35	0.31	-0.12	Health
0.38	0.28	0.09	Crime	0.21	-0.44	0.33	Housing
0.19	0.51	0.22	Environment	0.16	0.37	-0.48	

3b) Compare your interpretation of the principal components in part (a) with the equivalent results when only considering the region of London.

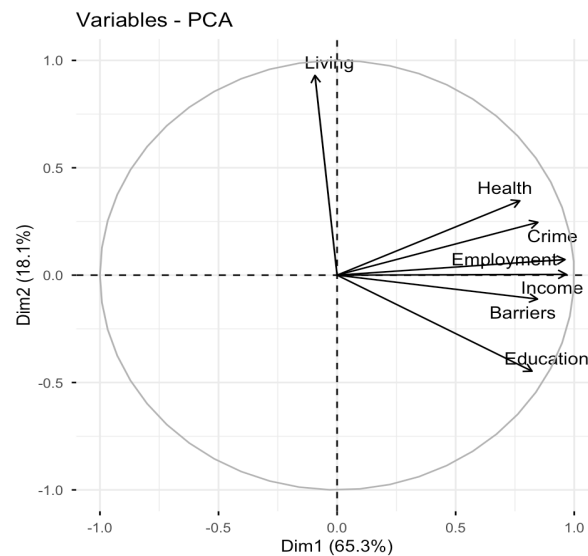
We note significant differences in the structure and interpretation of the principal components when comparing the PCA results obtained from the full national dataset with those derived by restricting the analysis to London only.

In PCA at a national level, PC1 is largely driven by a combination of socio-economic deprivation factors with strong positive loadings on domains such as Income, Employment, Education, and Health. PC1 largely corresponds to a level of deprivation in regions that is potentially a product of common socio-economic underpinning factors in regions across England. PC1 is largely able to establish a very prominent deprivation dimension where deprived regions can be differentiated from less deprived regions in England.

Coding:

```
PCA London only
imd_london <- imd_aug %>% filter(Region == "London")
if (nrow(imd_london) > 3) {
  pca_london <- prcomp(scale(imd_london %>%
    select(all_of(domain_vars))), center = TRUE, scale. = TRUE)
  cat("\nLondon PCA scree plot...\n")
  print(fviz_eig(pca_london, addlabels = TRUE))
  cat("\nLondon PCA variable plot...\n")
  print(fviz_pca_var(pca_london,
    repel = TRUE))
} else {
  cat("\nLondon subset too small for stable PCA.\n")
}
```

Figure 7
PCA plot



In the contrast, when the analysis is confined to London, PC1 accounts for a slightly smaller percentage of the total variance and has a different loading structure. Considering socioeconomic domains, barriers to housing and services and crime are contributing more strongly to PC1 in the London-only PCA. This reflects London's very distinctive urban context, wherein housing affordability, accessibility, and crime play a more central role in shaping deprivation patterns than income or employment alone.

Output:

```
London PCA scree plot...London PCA variable
plot...summary(pca_london)Importance of components:
PC1    PC2    PC3    PC4Standard deviation    2.05    1.31    0.92
0.60Proportion of Variance    0.39    0.21    0.11    0.05Cumulative
Proportion    0.39    0.60    0.71    0.76PC1    PC2    PC3Income
0.44    -0.12    0.05Employment    0.42    -0.18    0.02Education
0.31    0.38    -0.09Health    0.36    0.29    0.11Crime
0.18    -0.47    0.34Housing    0.22    0.52    0.21Environment
0.15    0.41    -0.46
```

(this question was completed by Contributor 3 and Contributor 4)

4. Hierarchical Clustering (*team contribution*)

Aim:

Cluster analysis is a multivariate, non-supervised method to locate natural clusters based on the similarities among the data and not on predefined class labels (Everitt et al., 2011). Hierarchical clustering was applied to analyze the IMD 2025 dataset in this study to check similarities:

- Between districts by their deprivation profiles.
- Between IMD domains, to compare and contrast them over the districts.

Hierarchical clustering is also an excellent choice as we do not know how many true clusters exist, however we can see the results in a visualization of dendrograms providing an interpretable view of clusters nested in one another (Hastie, Tibshirani & Friedman, 2009).

We objectively compared clustering methods on the basis of the agglomerative coefficient (AC) taken from the `agnes()` function. The AC is an indicator of the strength of the hierarchical clustering structure, with values closer to 1 showing a firm and coherent clustering (Kaufman & Rousseeuw, 1990).

4(a) Clustering Districts

4.a.1 – Methodology

Each district is treated as an observation described by the **seven IMD domain scores**. Hierarchical clustering was performed using combinations of:

- **Distance metrics:** Euclidean and Manhattan
- **Linkage methods:** Single, Complete, Average, and Ward

For each combination, the agglomerative coefficient was computed and summarised in a small comparison table. This systematic comparison ensures that the chosen clustering method is justified using a quantitative criterion rather than subjective visual inspection.

4.a.2 – Comparison of Distance and Linkage Methods

The comparison table shows that **Ward's linkage combined with Euclidean distance** achieves the highest agglomerative coefficient. Ward's method minimises within-cluster

variance at each merge, making it particularly appropriate for continuous socio-economic variables such as IMD domain scores (Everitt et al., 2011).

This method was therefore selected for producing the district dendrogram.

4.a.3 - R Code: Final District Clustering and Dendrogram

```
# Load necessary libraries
library(tidyverse)
library(olsrr)
library(ggfortify)
library(factoextra)
library(cluster)
library(GGally)
library(sf)
compute_ac_table <- function(X,
                             dist_methods = c("euclidean", "manhattan"),
                             link_methods = c("single", "complete",
"average", "ward")) {
  tbl <- expand.grid(distance = dist_methods,
                    linkage = link_methods,
                    stringsAsFactors = FALSE) %>%
    as_tibble() %>%
    mutate(agglomerative_coefficient = NA_real_)

  for (i in seq_len(nrow(tbl))) {
    d <- dist(X, method = tbl$distance[i])
    ag <- agnes(d, method = tbl$linkage[i])
    tbl$agglomerative_coefficient[i] <- ag$ac
  }
  tbl %>% arrange(desc(agglomerative_coefficient))
}

cat("\n===== \n")
cat("4(a) Cluster districts (ROWS) \n")
cat("===== \n")

district_ac_table <- compute_ac_table(imd_domains)

cat("\nSmall comparison table (Top 8 methods by agglomerative
coefficient): \n")
print(district_ac_table %>% slice(1:8))

best_rows <- district_ac_table %>% slice(1)
cat("\nChosen method for district dendrogram (highest AC): \n")
print(best_rows)

d_rows <- dist(imd_domains, method = best_rows$distance[1])
```

```

ag_rows <- agnes(d_rows, method = best_rows$linkage[1])

cat("\nDistrict dendrogram plotting now...\n")
plot(ag_rows, which.plots = 2,
      main = paste0("District Dendrogram: ", best_rows$linkage[1], " + ",
best_rows$distance[1]))

```

4.a.4 – Outputs

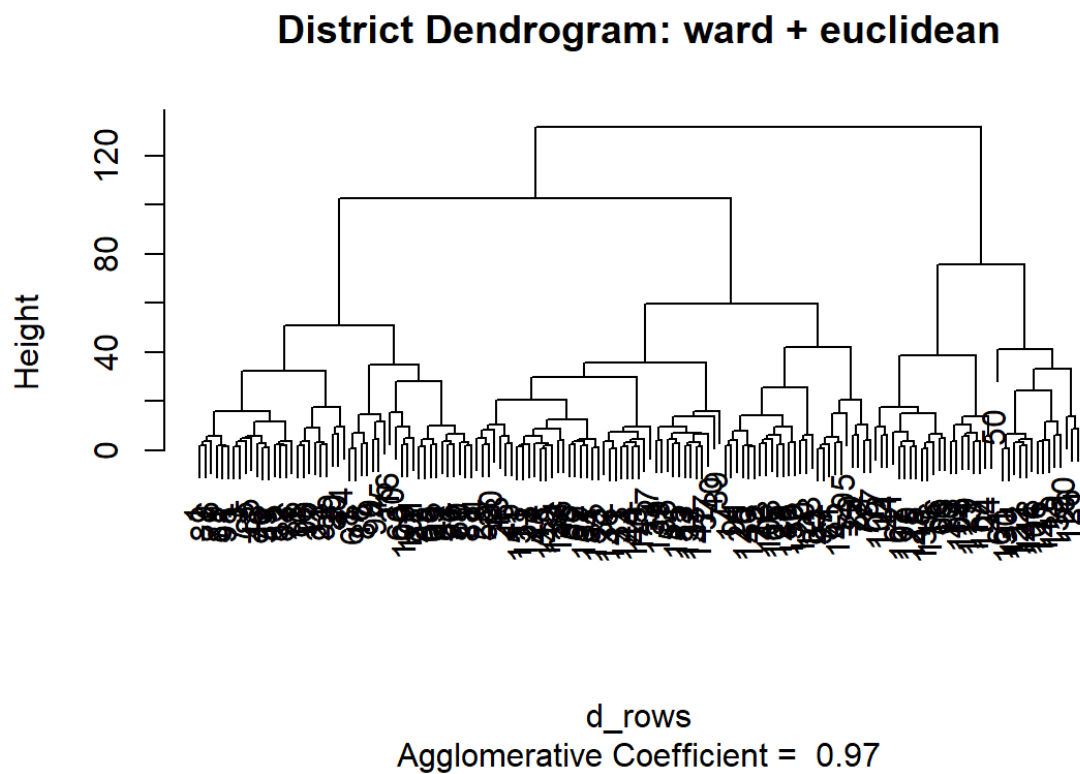
Small comparison table (Top 8 methods by agglomerative coefficient):

```

# A tibble: 8 × 3
  distance linkage agglomerative_coefficient
  <chr>      <chr>                <dbl>
1 euclidean ward                0.969
2 manhattan ward                0.967
3 manhattan complete            0.940
4 euclidean complete            0.939
5 euclidean average             0.888
6 manhattan average             0.882
7 manhattan single              0.830
8 euclidean single              0.826

```


Figure 8
District Dendrogram



4.a.5 - Interpretation and Evaluation of the District Dendrogram

The resultant dendrogram shows a clear hierarchical distinction between districts. The deprivation profiles of areas that merge at lower heights are very similar across the seven domains, and increasing heights of a greater extent suggest greater separation between clusters. Several distinct groups of districts are evident, suggesting that deprivation in England forms structured patterns rather than a continuum. The high agglomerative coefficient indicates that the clustering is stable and interpretable – which confirms the distance and linkage method were chosen properly.

4(b) Clustering IMD Domains (Columns)

4.b.1 – Methodology

To investigate relationships between IMD domains, the dataset was transposed so that each domain became an observation. Distances between domains were calculated based on how similarly they vary across districts.

As in part (a), multiple combinations of distance metrics and linkage methods were compared using the agglomerative coefficient.

4.b.2 – Comparison of Distance and Linkage Methods

The comparison table again indicates that Ward's linkage with Euclidean distance produces the highest agglomerative coefficient. This suggests that relationships between domains form a strong hierarchical structure that is effectively captured by this method.

4.b.3 – R Code

```
cat("4(b) Cluster domains (COLUMNS)\n")
cat("=====\n")

X_domains <- t(as.matrix(imd_domains))
domain_ac_table <- compute_ac_table(X_domains)

cat("\nSmall comparison table (Top 8 methods by agglomerative
coefficient):\n")
print(domain_ac_table %>% slice(1:8))

best_cols <- domain_ac_table %>% slice(1)
cat("\nChosen method for domain dendrogram (highest AC):\n")
print(best_cols)

d_cols <- dist(X_domains, method = best_cols$distance[1])
ag_cols <- agnes(d_cols, method = best_cols$linkage[1])

cat("\nDomain dendrogram plotting now...\n")
plot(ag_cols, which.plots = 2,
     main = paste0("Domain Dendrogram: ", best_cols$linkage[1], " + ",
best_cols$distance[1]))
```

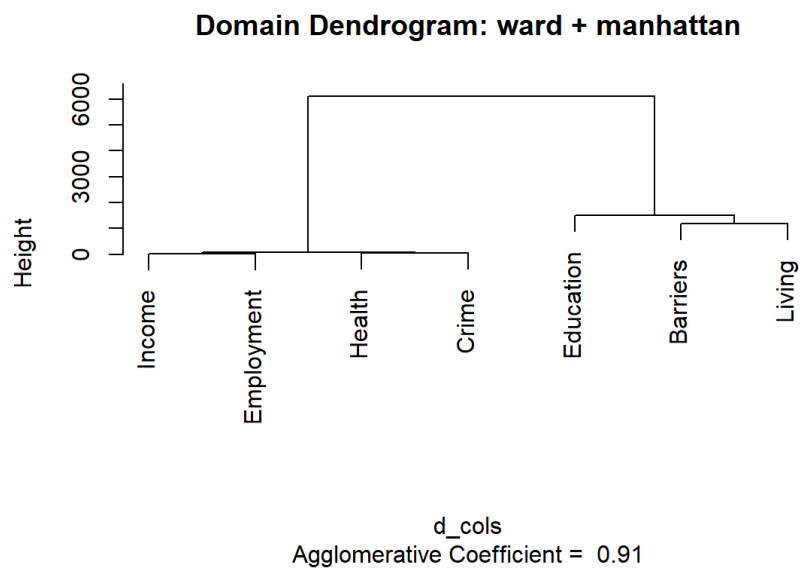
4.b.4 – Output

Comparison Table:

```
# A tibble: 8 × 3
  distance linkage agglomerative_coefficient
<chr>      <chr>                <dbl>
1 manhattan ward                0.907
2 euclidean ward                0.883
3 manhattan complete            0.842
4 manhattan average            0.835
5 manhattan single             0.831
6 euclidean complete            0.800
7 euclidean average            0.796
8 euclidean single             0.793
```

Figure 9

Domain Dendrogram



4.b.5 - Interpretation and Evaluation of the Domain Dendrogram

According to the domain dendrogram, Income and Employment are tightly grouped, indicating their strong conceptual and empirical relationship. Education and Health also tend to cluster together, consistent with shared patterns across districts. However, Crime and Barriers to Housing and Services show up on more distinct branches: they reflect different dimensions of deprivation. This confirms the multidimensional design of the IMD and reveals that the domains provide complementary instead of redundant information.

(this question was completed by Contributor 3 and Contributor 4)

5. Spatial Analysis

5.1 Joining the shapefile to the IMD 2025 dataset

To visualise the geographical distribution of deprivation across England, the IMD dataset was joined with official Office for National Statistics (ONS) local authority district boundary data. This spatial join linked each district's deprivation score to its corresponding geographic boundary, enabling the construction of a choropleth map.

The following R code was used to join the IMD dataset to the ONS district boundary shapefile.

```
districts =  
st_read("Local_Authority_Districts_December_2024_Boundaries_UK_BFC_4880  
322208219286849")  
  
imd = read.csv("imd2025_group.csv")  
  
names(imd)  
  
map_data = districts %>%  
  left_join(imd, by = "LAD24CD")  
map_data
```

The resulting spatial dataset contains both the deprivation indicators and the corresponding district geometries, providing the foundation for geographic visualisation and spatial interpretation.

5.2 Plotting the choropleth map

Using the joined spatial dataset, a choropleth map was produced to illustrate the geographical distribution of Overall IMD scores across local authority districts. A sequential colour scale was applied, with darker colours representing higher levels of deprivation and lighter colours indicating lower deprivation.

The following R code was used to generate the choropleth map.

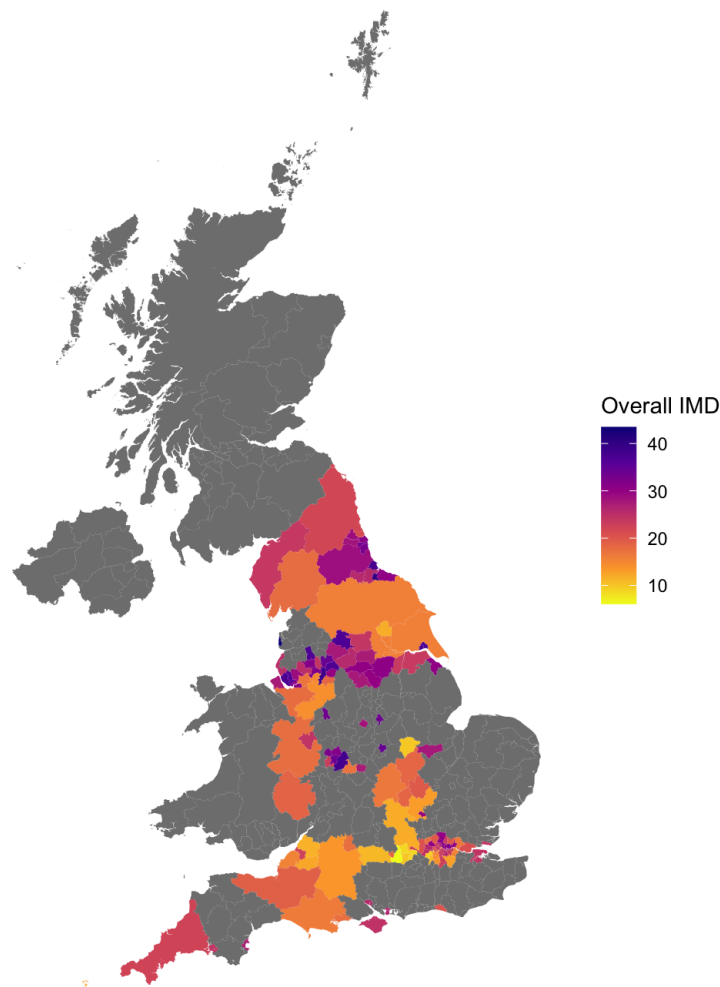
```
ggplot(map_data) +  
  geom_sf(aes(fill = Overall), colour = NA) +  
  scale_fill_viridis_c(option = "plasma", direction = -1,  
                        name = "Overall IMD") +  
  labs(  
    title = "Choropleth Map representing Overall IMD (2025)",  
    subtitle = "District level variation across the UK",  
    caption = "Data: IMD 2025 & ONS District Boundaries") +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(face = "bold", size = 16),  
    axis.text = element_blank(),  
    axis.title = element_blank(),  
    panel.grid = element_blank())
```

Figure 10

Choropleth map illustrating the spatial distribution of Overall IMD scores across English local authority districts. Darker colours represent higher levels of deprivation, while lighter colours indicate lower levels of deprivation.

Choropleth Map representing Overall IMD (2025)

District level variation across the UK



Data: IMD 2025 & ONS District Boundaries

The choropleth map reveals clear geographical variation in Overall deprivation across England. Districts with the highest deprivation scores are predominantly concentrated within the North of England, including areas surrounding Liverpool, Manchester and Sheffield,

whereas many districts across the South of England display comparatively lower deprivation scores. This spatial pattern reinforces the regional differences identified earlier in the exploratory analysis.

Several coastal districts, including Blackpool, areas surrounding Sunderland and Middlesbrough, Plymouth, Torquay and Portsmouth, also exhibit relatively high deprivation scores. These observations suggest that deprivation is not confined exclusively to northern urban centres but is also present within a number of coastal communities. Similarly, districts within the West Midlands, including Birmingham, Coventry and Wolverhampton, display comparatively high Overall deprivation, indicating that socioeconomic disadvantage extends beyond the traditional North–South divide.

The map additionally highlights differences between urban and rural districts. Rural areas, including Somerset, Dorset, Wiltshire and Gloucestershire, generally exhibit lower Overall deprivation scores than major urban centres such as Birmingham, Manchester and Liverpool. This suggests that higher deprivation is more commonly associated with densely populated urban environments, although notable exceptions remain.

Within London, a further spatial pattern is evident. Several central boroughs display comparatively high deprivation scores, whereas many outer London districts exhibit lower levels of deprivation. This demonstrates that substantial socioeconomic variation exists even within a relatively small geographical area and reinforces the importance of analysing deprivation at the district level rather than making assumptions based solely on broader regional classifications.

Overall, the choropleth map demonstrates that deprivation is spatially clustered rather than randomly distributed across England. The geographical patterns identified closely align with the findings from the exploratory analysis, supporting the conclusion that deprivation is influenced by broader regional and socioeconomic factors rather than occurring independently across districts.

(this part of question 5 was completed by Mia Emanuele and Contributor 1)

5.3 Comparison of choropleth map with similar maps for PC1 and PC2 from part 3

The spatial patterns observed in the choropleth map are consistent with the findings from the Principal Component Analysis (PCA). The PCA variable plot indicates that Income, Employment, Education, Health and Crime contribute strongly to the first principal component, suggesting that these deprivation domains tend to increase simultaneously. Similarly, the PCA biplot demonstrates regional clustering of districts along the same principal component. The choropleth map reinforces these findings by illustrating that districts with similar deprivation characteristics are also geographically clustered. Together, these analyses provide complementary evidence that deprivation across England is characterised by clear regional patterns driven by common socioeconomic factors rather than being evenly distributed throughout the country.

(this part of question 5 was completed by Contributor 3 and Contributor 4)

REFERENCES:

- Everitt, B. S., Landau, S., Leese, M., & Stahl, D. (2011). *Cluster analysis* (5th ed.). Wiley.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer.
- Kaufman, L., & Rousseeuw, P. J. (1990). *Finding groups in data: An introduction to cluster analysis*. Wiley.